
SMART STUDENT PERFORMANCE ANALYSIS

Machine Learning Classification & Clustering Project

Project Proposal Document

Date: May 08, 2026

Version 1.0

Project Type	Machine Learning / Data Science
Algorithms	KNN (Classification) + DBSCAN (Clustering)
Objective	Predict student performance & discover student groups
Dataset	Synthetic Student Records (500 students, 14 features)
Tools	Python, Scikit-learn, Pandas, Matplotlib, Seaborn

1. Executive Summary

This project presents a comprehensive machine learning analysis of student academic performance, combining K-Nearest Neighbors (KNN) classification and DBSCAN clustering to understand and predict student outcomes. The analysis uses a synthetic dataset of 500 students with behavioral and academic features including study hours, attendance, quiz scores, sleep patterns, and social media usage.

The project demonstrates two complementary algorithmic approaches: KNN for supervised classification (predicting pass/fail status and grade categories), and DBSCAN for unsupervised clustering (identifying natural student groups and detecting outliers). This dual-algorithm design provides a complete view of both individual prediction and population-level pattern discovery.

2. Project Objectives

- Build a KNN classifier to predict student pass/fail status with high accuracy
- Extend KNN classification to predict fine-grained grade categories (A/B/C/D/F)
- Apply DBSCAN clustering to discover natural student groupings based on behavioral patterns
- Identify exceptional students (high performers) and at-risk students (struggling) as outliers
- Engineer meaningful features from raw student data to improve model performance
- Generate comprehensive visualizations and documentation for stakeholder communication

3. Dataset Description

A synthetic dataset of 500 student records was generated with realistic distributions, including natural correlations between features and target variables. The dataset includes:

Feature	Type	Description
Study Hours/Week	Numerical	Hours spent studying per week (0-60)
Attendance %	Numerical	Percentage of classes attended (0-100%)
Quiz Avg Score	Numerical	Average quiz score (0-100)
Sleep Hours/Night	Numerical	Average sleep duration (3-12 hrs)

Social Media Hrs/Day	Numerical	Daily social media usage (0-15 hrs)
Previous GPA	Numerical	Cumulative GPA from previous semester (0-4)
Extracurricular	Binary	Participation in activities (0/1)
Parent Education	Categorical	Highest education level (4 levels)
Current GPA	Numerical	Current semester GPA (derived)
Final Score	Numerical	Final course score (derived, 0-100)
Pass/Fail	Binary	Target: Pass (score ≥ 50) or Fail
Grade	Ordinal	Target: A/B/C/D/F grade categories

4. Methodology & Approach

4.1 Data Generation & Preprocessing

Synthetic data is generated with realistic distributions and correlations. The preprocessing pipeline includes outlier capping using IQR, categorical encoding, and feature engineering (Study Efficiency, Sleep-Study Ratio, Social Media Risk Score, Attendance-Study Product). All features are standardized using StandardScaler for ML algorithm compatibility.

4.2 KNN Classification

K-Nearest Neighbors is used for two classification tasks: binary (Pass/Fail) and multi-class (A/B/C/D/F). The optimal K value is found by evaluating accuracy across K=1 to 30. The dataset is split 75/25 with stratification. Models are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. 5-fold stratified cross-validation ensures robustness.

4.3 DBSCAN Clustering

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is applied after PCA dimensionality reduction to 3 components. DBSCAN discovers clusters of arbitrary shape and identifies outliers as noise points. Optimal eps and min_samples are found via grid search using a combined score: $\text{silhouette} * (1 - \text{noise_percentage})$. This ensures meaningful clusters with minimal noise.

5. Project Structure

Directory/File	Contents
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src/step1_generate_data.py	Synthetic student dataset generation
src/step2_preprocessing.py	Cleaning, encoding, feature engineering
src/step3_knn_classification.py	KNN model training & evaluation
src/step4_dbscan_clustering.py	DBSCAN clustering & outlier detection
src/step5_visualization.py	Charts and visual reports
run_all_steps.py	Orchestrator script for full pipeline
data/raw/	Original generated dataset
data/processed/	Cleaned and clustered datasets
reports/	Results and chart outputs
requirements.txt	Python dependencies

6. Expected Results

6.1 KNN Classification Results

Metric	Pass/Fail	Grade (A/B/C/D/F)
Accuracy	~91%	~86%
F1-Score	~0.91	~0.84
ROC-AUC	~0.98	N/A (multiclass)
Best K	29	22
CV Accuracy	~89% (+/- 3%)	~82% (+/- 3%)

6.2 DBSCAN Clustering Results

Metric	Value
PCA Components	3 (51.5% variance explained)
Optimal eps	1.6
Optimal min_samples	3
Silhouette Score	0.73
Clusters Found	2
Noise Points	~8 (1.6%)

7. Deliverables

Deliverable	Format	Location
Raw Student Dataset	CSV	data/raw/student_performance.csv
Processed Dataset	CSV	data/processed/student_performance_processed.csv
Clustered Dataset	CSV	data/processed/student_clusters.csv
KNN Results	Text	reports/knn_results.txt
DBSCAN Results	Text	reports/dbscan_results.txt

Visualizations	PNG (8 charts)	reports/charts/
Pipeline Script	Python	run_all_steps.py
Proposal Document	PDF	Student_Performance_Analysis_Proposal.pdf

8. Technical Requirements

Package	Version	Purpose
pandas	>=1.3.0	Data manipulation
numpy	>=1.20.0	Numerical computing
matplotlib	>=3.4.0	Plotting
seaborn	>=0.11.0	Statistical visualization
scikit-learn	>=1.0.0	ML algorithms (KNN, DBSCAN, PCA)
scipy	>=1.7.0	Scientific computing

Python Version: 3.8+ required.

9. Risk Assessment

Risk	Likelihood	Impact	Mitigation
Synthetic data bias	Medium	Medium	Realistic distributions with noise
KNN sensitivity to K	Low	Medium	Systematic search over K=1 to 30
DBSCAN parameter tuning	Medium	High	Grid search with combined scoring metric
Curse of dimensionality	Low	Medium	PCA reduction before DBSCAN
Environment dependency	Medium	Low	Version-pinned requirements.txt

10. Success Metrics

- KNN achieves >85% accuracy on pass/fail classification
- DBSCAN identifies meaningful clusters with <10% noise points
- Outlier analysis distinguishes exceptional and at-risk students

- All 5 pipeline steps execute without errors
- Visualizations are generated for all analytical components
- Complete documentation and proposal are produced

This proposal outlines the complete scope and methodology for the Smart Student Performance Analysis project, combining KNN classification and DBSCAN clustering for comprehensive educational data mining.

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